Customer Purchase Behavior Analysis & Prediction for Amazon

Machine Learning & Business Intelligence Project

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1. Problem Statement

- Improve customer retention and revenue forecasting using data-driven insights.
- Identify high-value customers and churn risk.
- Enable targeted marketing and personalization strategies.

2. Dataset Overview

- Customer demographics and purchase history.
- Loyalty indicators, discount usage, and payment methods.
- Cleaned and ML-ready dataset used for modeling and dashboard creation.

3. Data Cleaning & Preprocessing

- Handled missing values using median and mode imputation.
- Verified absence of duplicate customer records.
- Ensured consistent data types and analyzed outliers.

4. Feature Engineering

- Validated Loyalty Score for customer value assessment.
- Used Discount Applied and Return Status indicators.
- Leveraged Customer Segment and Preferred Shopping Channel features.

5. Task 1: Customer Segmentation (K-Means Clustering)

- Aggregated data at customer level.
- Features used: total spend, purchase frequency, loyalty score.
- Applied Elbow Method to determine optimal clusters.
- Segmented customers into Low Value, Regular, and High Value (VIP).

6. Task 2: Customer Lifetime Value Prediction (Linear Regression)

- Target variable: Customer Lifetime Value (CLV).
- Predictors: age, purchase amount, loyalty score, discount usage, payment method.
- Model evaluated using R^2, MAE, and RMSE.
- Identified key drivers influencing long-term customer value.

7. Task 3: Customer Churn Prediction (Logistic Regression)

- Target variable: Churn (0 = Active, 1 = Churned).
- Used demographic and behavioral predictors.
- Evaluated using classification metrics.
- Identified factors contributing to customer churn.

8. Power BI Dashboard Overview

- Executive Overview with key KPIs.
- Customer Segmentation analysis.
- CLV and Revenue insights.
- Churn analysis with interactive slicers.

9. Key Insights & Business Recommendations

- Loyalty score strongly influences CLV and churn.
- High-value customers contribute significantly to long-term revenue.
- Discount-driven customers exhibit higher churn risk.
- Recommend targeted loyalty programs and personalized engagement.

10. Limitations & Future Scope

- Analysis based on historical data and proxy churn indicators.
- Future scope includes time-based features.
- Advanced ML models can further improve predictive performance.